

COMMUNICATION

1. Iron is always heavier than wood.
Explain whether the statement is true or false.

No, if the iron has the same volume this is true. [2 marks]
But it could be either, if the wood has more volume, its weight will surpass that of iron. It depends on the volume of each.

2. Describe how an application such as hydraulic brake fluid system in a car basically works using a principal learned in this unit.

Since it is a closed loop the pressure applied at one end is applied everywhere so the brake cylinders also move and apply a force to the disk which causes them to stop the car. [3 marks]
See solution! (2 marks)
Master cylinder, Pedal, Brake cylinder, Brake pads, Disk attached to wheel. (1/2)

3. Explain how the tube in the diagram, known as a siphon, can transfer liquid from one container to a lower one even though the liquid must flow uphill for part of its journey. (Note that the tube must be filled with liquid to start with.)

Since the tube is filled, the water in the tube has a gravitational force that pulls the liquid up. So when it's filled the weight of the water pulls the water up the other side. [3 marks]
See solution! (2 marks)
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z, AA, AB, AC, AD, AE, AF, AG, AH, AI, AJ, AK, AL, AM, AN, AO, AP, AQ, AR, AS, AT, AU, AV, AW, AX, AY, AZ, BA, BB, BC, BD, BE, BF, BG, BH, BI, BJ, BK, BL, BM, BN, BO, BP, BQ, BR, BS, BT, BU, BV, BW, BX, BY, BZ, CA, CB, CC, CD, CE, CF, CG, CH, CI, CJ, CK, CL, CM, CN, CO, CP, CQ, CR, CS, CT, CU, CV, CW, CX, CY, CZ, DA, DB, DC, DD, DE, DF, DG, DH, DI, DJ, DK, DL, DM, DN, DO, DP, DQ, DR, DS, DT, DU, DV, DW, DX, DY, DZ, EA, EB, EC, ED, EE, EF, EG, EH, EI, EJ, EK, EL, EM, EN, EO, EP, EQ, ER, ES, ET, EU, EV, EW, EX, EY, EZ, FA, FB, FC, FD, FE, FF, FG, FH, FI, FJ, FK, FL, FM, FN, FO, FP, FQ, FR, FS, FT, FU, FV, FW, FX, FY, FZ, GA, GB, GC, GD, GE, GF, GG, GH, GI, GJ, GK, GL, GM, GN, GO, GP, GQ, GR, GS, GT, GU, GV, GW, GX, GY, GZ, HA, HB, HC, HD, HE, HF, HG, HH, HI, HJ, HK, HL, HM, HN, HO, HP, HQ, HR, HS, HT, HU, HV, HW, HX, HY, HZ, IA, IB, IC, ID, IE, IF, IG, IH, II, IJ, IK, IL, IM, IN, IO, IP, IQ, IR, IS, IT, IU, IV, IW, IX, IY, IZ, JA, JB, JC, JD, JE, JF, JG, JH, JI, JJ, JK, JL, JM, JN, JO, JP, JQ, JR, JS, JT, JU, JV, JW, JX, JY, JZ, KA, KB, KC, KD, KE, KF, KG, KH, KI, KJ, KK, KL, KM, KN, KO, KP, KQ, KR, KS, KT, KU, KV, KW, KX, KY, KZ, LA, LB, LC, LD, LE, LF, LG, LH, LI, LJ, LK, LL, LM, LN, LO, LP, LQ, LR, LS, LT, LU, LV, LW, LX, LY, LZ, MA, MB, MC, MD, ME, MF, MG, MH, MI, MJ, MK, ML, MM, MN, MO, MP, MQ, MR, MS, MT, MU, MV, MW, MX, MY, MZ, NA, NB, NC, ND, NE, NF, NG, NH, NI, NJ, NK, NL, NM, NN, NO, NP, NQ, NR, NS, NT, NU, NV, NW, NX, NY, NZ, OA, OB, OC, OD, OE, OF, OG, OH, OI, OJ, OK, OL, OM, ON, OO, OP, OQ, OR, OS, OT, OU, OV, OW, OX, OY, OZ, PA, PB, PC, PD, PE, PF, PG, PH, PI, PJ, PK, PL, PM, PN, PO, PP, PQ, PR, PS, PT, PU, PV, PW, PX, PY, PZ, QA, QB, QC, QD, QE, QF, QG, QH, QI, QJ, QK, QL, QM, QN, QO, QP, QQ, QR, QS, QT, QU, QV, QW, QX, QY, QZ, RA, RB, RC, RD, RE, RF, RG, RH, RI, RJ, RK, RL, RM, RN, RO, RP, RQ, RR, RS, RT, RU, RV, RW, RX, RY, RZ, SA, SB, SC, SD, SE, SF, SG, SH, SI, SJ, SK, SL, SM, SN, SO, SP, SQ, SR, SS, ST, SU, SV, SW, SX, SY, SZ, TA, TB, TC, TD, TE, TF, TG, TH, TI, TJ, TK, TL, TM, TN, TO, TP, TQ, TR, TS, TT, TU, TV, TW, TX, TY, TZ, UA, UB, UC, UD, UE, UF, UG, UH, UI, UJ, UK, UL, UM, UN, UO, UP, UQ, UR, US, UT, UU, UV, UW, UX, UY, UZ, VA, VB, VC, VD, VE, VF, VG, VH, VI, VJ, VK, VL, VM, VN, VO, VP, VQ, VR, VS, VT, VU, VV, VW, VX, VY, VZ, WA, WB, WC, WD, WE, WF, WG, WH, WI, WJ, WK, WL, WM, WN, WO, WP, WQ, WR, WS, WT, WU, WV, WW, WX, WY, WZ, XA, XB, XC, XD, XE, XF, XG, XH, XI, XJ, XK, XL, XM, XN, XO, XP, XQ, XR, XS, XT, XU, XV, XW, XX, XY, XZ, YA, YB, YC, YD, YE, YF, YG, YH, YI, YJ, YK, YL, YM, YN, YO, YP, YQ, YR, YS, YT, YU, YV, YW, YX, YY, YZ, ZA, ZB, ZC, ZD, ZE, ZF, ZG, ZH, ZI, ZJ, ZK, ZL, ZM, ZN, ZO, ZP, ZQ, ZR, ZS, ZT, ZU, ZV, ZW, ZX, ZY, ZZ.

Communication marks are awarded throughout the test according to the shown rubric. [2 marks]
need to further explain why

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly. [1/2]	Some mathematical/physical conventions are used correctly. [1]	Most mathematical/physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/physical conventions correctly. [2]

Mass Densities of Common Substances

Substance	Mass Density ρ (kg/m ³)	Substance	Mass Density ρ (kg/m ³)
Solids		Liquids	
Aluminum	2700	Blood (whole, 37 °C)	1060
Brass	8470	Ethyl alcohol	806
Concrete	2200	Mercury	13 600
Copper	8890	Oil (hydraulic)	800
Diamond	3520	Water (4 °C)	1.000×10^3
Gold	19 300	Gases	
Ice	917	Air	1.29
Iron (steel)	7860	Carbon dioxide	1.98
Lead	11 300	Helium	0.179
Quartz	2660	Hydrogen	0.0899
Silver	10 500	Nitrogen	1.25
Wood (yellow pine)	550	Oxygen	1.43

THINKING/INQUIRY

1. A large rectangular raft (density 650 kg/m^3) is floating on a lake. The surface area of the top of the raft is 8.2 m^2 and its volume is 1.80 m^3 . The density of the lake water is 1000 kg/m^3 . Show all your work.

- a) Calculate the height h of the portion of the raft that is above the surrounding water. Show all your work.

$V = A \times h$ Will float since density is less than water

$\rho_w \cdot g \cdot V = mg$

$\rho_{\text{water}} \cdot g \cdot V_{\text{submerged}} = \rho_{\text{wood}} \cdot V_{\text{total}} \cdot g$

$V_{\text{submerged}} = \frac{\rho_{\text{wood}} \cdot V_{\text{total}}}{\rho_{\text{water}}}$

$A \cdot h_{\text{sub}} = \frac{\rho_{\text{wood}} \cdot V_{\text{total}}}{\rho_{\text{water}}}$

$h_{\text{sub}} = \frac{\rho_{\text{wood}} \cdot V_{\text{total}}}{A \cdot \rho_{\text{water}}}$

$h_{\text{sub}} = \frac{650 \cdot 1.80}{8.2 \cdot 1000}$

$h_{\text{sub}} = 0.1427 \text{ m}$

Final answer: 0.1427 m

- b) Calculate the magnitude of the buoyant force on the raft and state its direction. Show all your work.

$F_B = \text{Weight of water displaced}$

$F_B = \rho_w \cdot V_{\text{submerged}} \cdot g$

$F_B = 1000 \cdot 0.1427 \cdot 9.8$

$F_B = 1394.6 \text{ N}$

Final answer: 1394.6 N

- c) If the average mass of a person is 75 kg , calculate the maximum number of people that can be on the raft without the top of the raft sinking below the surface of the water. (Assume that the people are evenly distributed on the raft.) Show all your work.

$\text{Max } B = \rho_w \cdot V_{\text{total}} \cdot g$

$\text{Max } B = 17640 \text{ N}$

$\text{Weight of Raft} = 1394.6 \text{ N}$

$17640 \text{ N} - 1394.6 \text{ N} = 16245.4 \text{ N}$

$16245.4 \text{ N} = m_{\text{people}} \cdot g$

$m_{\text{people}} = \frac{16245.4 \text{ N}}{9.8 \text{ m/s}^2}$

$m_{\text{people}} = 1657.6 \text{ kg}$

$\frac{1657.6 \text{ kg}}{75 \text{ kg/person}} = 22.1 \text{ people}$

Final answer: 22 people

Name: _____

Date: _____

2. Water and then oil (which don't mix) are poured into a U-shaped tube, open at both ends. They come to Equilibrium as shown in the diagram. What is the density of the oil? Show all your work.

[3 marks]

To be in equilibrium pressures have to be the same

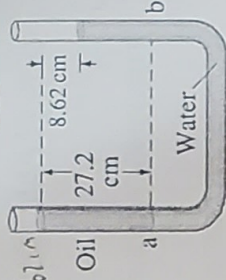
$$\rho_{oil} \cdot h_{oil} \cdot g = \rho_{water} \cdot h_{water} \cdot g$$

$$\rho_{oil} = \frac{\rho_{water} \cdot h_{water}}{h_{oil}}$$

$$h_{water} = 27.2 \text{ m} - 8.62 \text{ m} = 18.58 \text{ m}$$

$$h_{oil} = 0.1858 \text{ m}$$

$$h_{oil} = 0.272 \text{ m}$$



$$\rho_{oil} = \frac{(1000 \text{ kg/m}^3)(0.1858 \text{ m})}{(0.272 \text{ m})}$$

$$\rho_{oil} = 683.082 \text{ kg/m}^3$$

Final answer: 683 kg/m³

3. A 15.875 mm diameter garden hose is used to fill a round swimming pool 6.1 m in diameter. How long will it take to fill the pool to a depth of 1.4 m if water flows from the hose at a speed of 0.40 m/s? Express your answer in days. Show all your work.

[4 marks]

$$\frac{15.875 \times 10^{-3}}{2} = R_{hose}$$

$$0.0079375 \text{ m} = R_{hose}$$

$$V_{pool} = \pi \left(\frac{d}{2}\right)^2 \cdot h$$

$$V_{pool} = 40.91453192 \text{ m}^3$$

$$V_{pool} = V_{pool} \cdot \text{flow rate}$$

$$T = \frac{40.91453}{0.40 \text{ m/s}} = 102.286325 \text{ s}$$

$$T = \frac{102.286325 \text{ s}}{86400 \text{ s/day}} = 0.001183 \text{ days}$$

$$T_{days} = \frac{1}{60 \times 60 \times 24} = 5.98 \times 10^{-5} \text{ days}$$

Final answer: 6 days

4. A 6.0 cm diameter horizontal pipe gradually narrows to 4.5 cm. When water flows through this pipe at a certain rate, the gauge pressure in these two sections is 33.5 kPa and 22.6 kPa, respectively. What is the volume rate of flow in meter cube per second (m³/s)? Show all your work.

[5 marks]

$$P_1 + \rho g h_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

$$2(P_1 - P_2) = \rho (v_2^2 - v_1^2)$$

$$\frac{2(P_1 - P_2)}{\rho} = v_2^2 - v_1^2 = \left(\frac{A_1}{A_2}\right)^2 v_1^2 - v_1^2 = v_1^2 \left(\left(\frac{A_1}{A_2}\right)^2 - 1\right)$$

Final answer: 0.015 m³/s

Name: _____

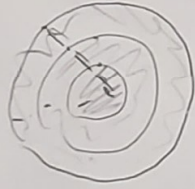
Date: _____

APPLICATION

1. The Earth is not a uniform sphere, but has regions of varying density. Consider a simple model of the Earth divided into three regions—inner core, outer core, and mantle. Each region is taken to have a unique constant density (the average density of that region in the real Earth):

Region	Radius (km)	Density (kg/m ³)
Inner Core	0 – 1220	13,000
Outer Core	1220 – 3480	11,100
Mantle	3480 – 6380	4400

- a) Use this model to calculate the average density of the entire three-layered Earth. Show all your work.



$$\rho = \frac{M}{V}$$

$$\frac{4\pi}{3} (R_{\text{mantle}}^3 - R_{\text{outer}}^3) = V_{\text{mantle}}$$

$$\frac{4\pi}{3} (R_{\text{outer}}^3 - R_{\text{inner}}^3) = V_{\text{outer}}$$

$$\frac{4\pi}{3} (R_{\text{inner}}^3) = V_{\text{inner}}$$

[5 marks]

Continued on scrap

Final answer: 5500 kg/m³

- b) If the radius of the Earth is 6380 km and its mass is 5.98×10^{24} kg, determine the actual average density of the Earth. Show all your work.

$$\rho = \frac{M}{V}$$

$$= \frac{5.98 \times 10^{24}}{\frac{4\pi}{3} (6380 \times 1000)^3}$$

$$\rho = \frac{5497.31393 \text{ kg/m}^3}{5500.38257}$$

$$\rho = 5497.31393 \text{ kg/m}^3$$

[1.5 mark]

Final answer: 5500 kg/m³

- c) Compare the actual average density of the Earth found from part b) with the one you determined in part a) as a percent difference. Show all your work.

$$\rho_{\text{act}} = \frac{5497.31393 \text{ kg/m}^3}{5500.38257}$$

$$\rho_{\text{act}} = 0.9994047655$$

$$1 - \rho_{\text{act}} = 0.05\% \text{ difference}$$

[1.5 mark]

Final answer: 0.05452%

2. In humans, blood flows from the heart into the aorta, from which it passes into the major arteries. The blood returns to the heart via the veins. The radius of the aorta is about 1.2 cm, and the blood passing through it has a speed of about 40 cm/s.

Calculate the average speed of blood flow in the major arteries of the body. Assume that the arteries have an average radius of 0.8 cm. Express the answer in meter per seconds (m/s). Show all your work.

$$\rho = 0.012 \text{ m}$$

$$0.4 \text{ m/s}$$

$$V \cdot A = V \cdot A$$

$$\frac{V \cdot A}{A}$$

$$= V_{\text{artery}}$$

$$0.4 \text{ m/s} = V$$

[2 marks]

Final answer: 0.9 m/s